# Towards Large-scale Unified Recommender System

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**Tencent**

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Jeju, Korea

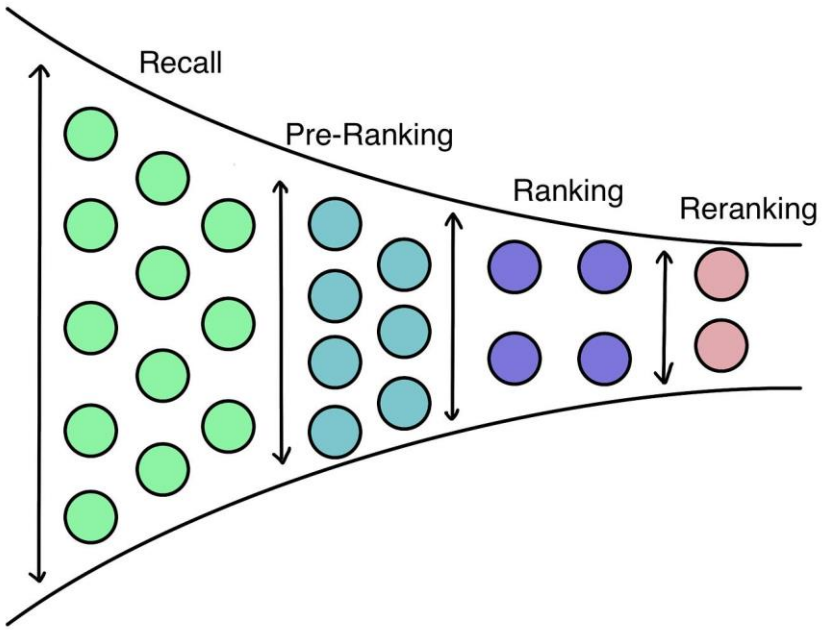
Tencent Marketing Solution

# Outline

- Overview of Large-scale Recommendation Systems
- Feature Interaction Recommenders
- Sequential Recommenders
- Unified Recommenders

# Overview of Industrial Recommender Systems

# Cascading Phases of Industrial RecSys




billions thousands hundreds dozens

xDeepFM
SIM
HeMix
SASRec
Wide & Deep
HSTU
TransAct
AFM
TIN
Wukong
GRU4Rec
Multi-Embedding
FM
FFM
AutoInt
NFM
BST
TWIN
RankMixer
DSIN
FwFM
DIEN
D-MoE
Feature Generation
DARE
HiFormer
CF
FinalMLP
PNN
DHEN
Wukong
BERT4Rec
DIN
Longer
DCN V2

# Recommendation Models

## Feature Interaction Models

Wide & Deep, FM, CF, FFM, FwFM, FmFM, NFM, AFM, DCN, DCNV2, xDeepFM, AutoInt, FinalMLP, RankMixer, Wukong, DHEN, Multi-Embedding, FeatureGeneration, HiFormer, HyFormer, OneTrans, InterFormer, HeMix, Kunlun, MixFormer, MtMixAtt, HHFT

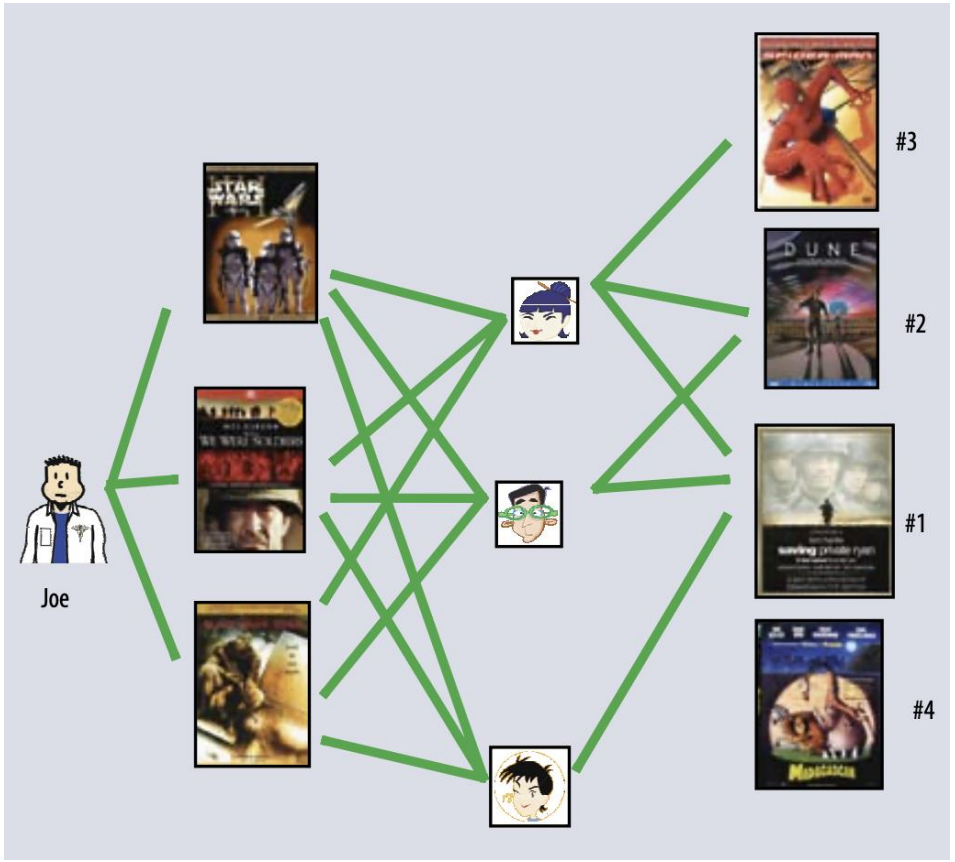
## Sequential Models

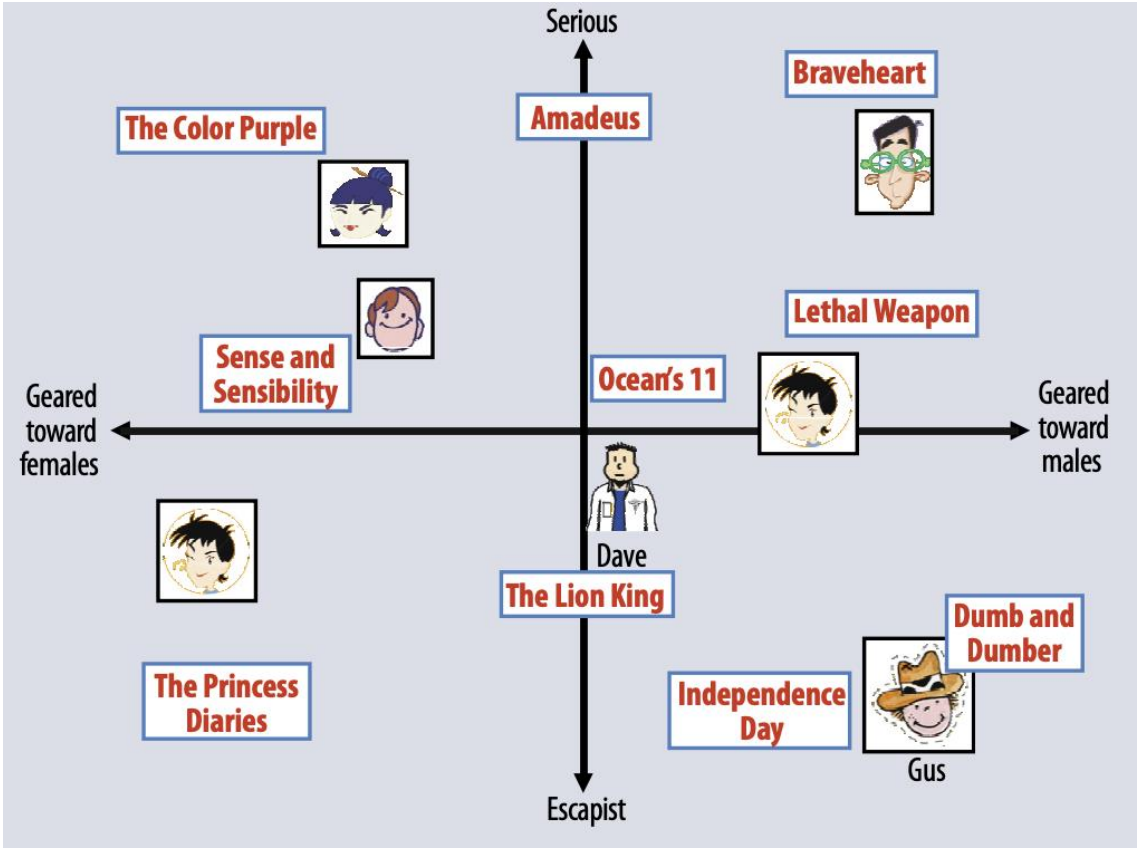
DIN, DIEN, DSIN, TIN, SIM, TWIN, TWINV2, DARE, Longer, SASRec, BST, BERT4Rec, GRU4Rec, HSTU, TransAct, Longer

# Feature Interaction Recommenders

# Matrix Factorization

- Factorize users and items to latent representations

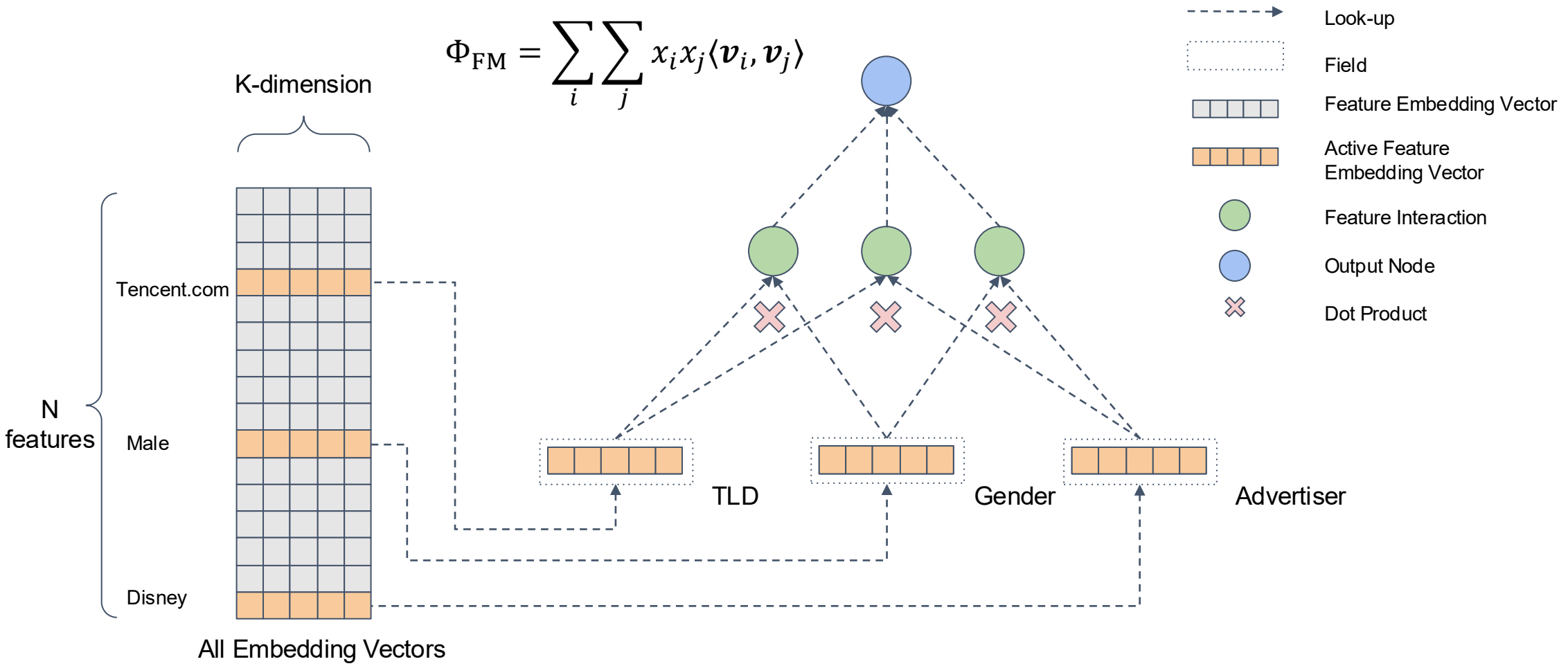



Matrix Factorization Techniques for Recommender Systems. 2009.

# Factorization Machines

$$\Phi_{\mathrm{FM}} = \sum_i \sum_j x_i x_j \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$$

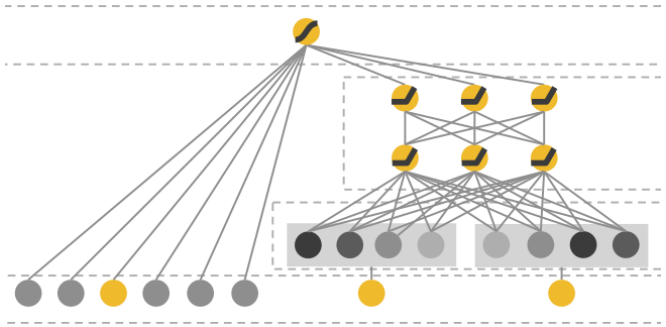


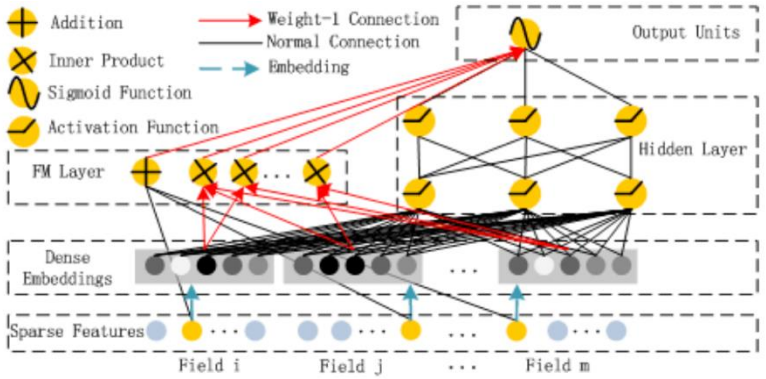

Factorization Machines. ICDM 2010.

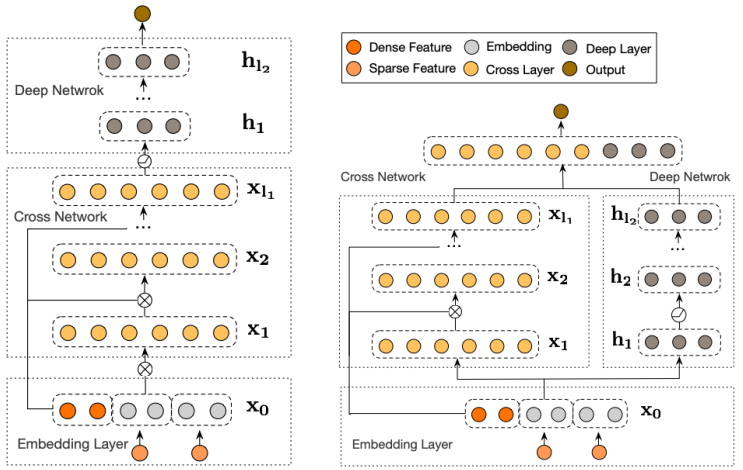
# Wide & Deep Family

Follow a Wide & Deep framework

- Wide: **explicit** interactions.
- Deep: **implicit** high-order interactions.






(a) Stacked (b) Parallel

Wide & Deep

- Wide: Logistic Regression
- Deep: DNN

DeepFM

- Wide: FM
- Deep: DNN

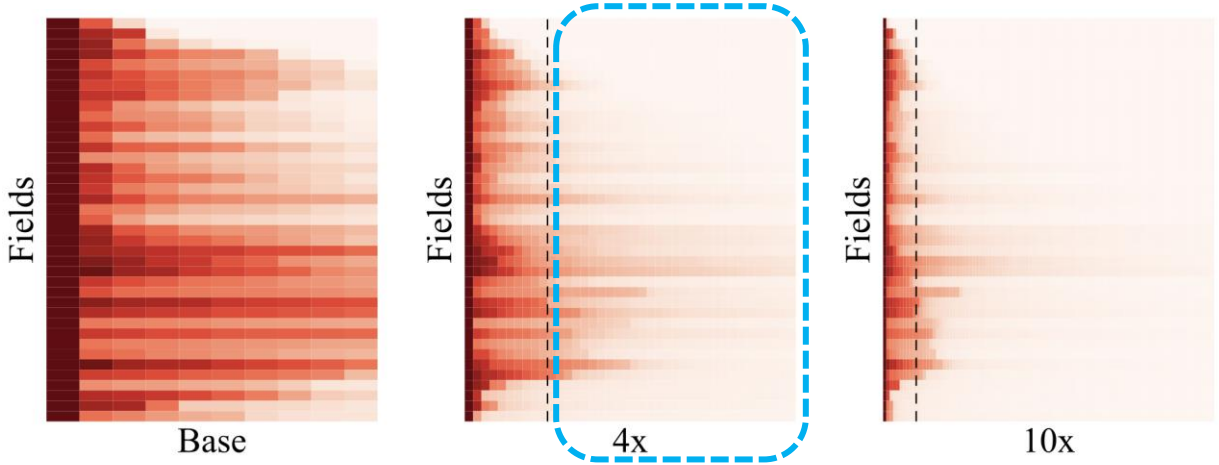
DCN V2

- Explicit: CrossNet
- Implicit: DNN.

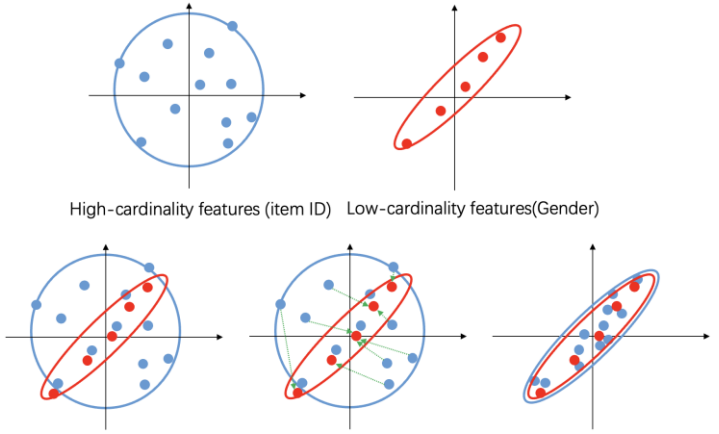

Wide & deep learning for recommender systems. 2016.
DeepFM: A Factorization-Machine based Neural Network for CTR Prediction. 2017.
DCN v2: Improved Deep & Cross Network and Practical Lessons for Web-scale Learning to Rank Systems. WWW 2021.

# Scaling Law in RecSys

# Dimensional Collapse




Collapsed dimensions




There are **many dimensionality-ill features** (e.g. low cardinality). Interaction with these features lead to the dimension collapse of the other features.

Assume A is a high-cardinality feature and B is a low-cardinality one.

$$v_i \in A, v_j \in B, W = \text{projection matrix}$$

$$\nabla_{v_i} l = (\sigma(s) - y)v_j + \lambda v_i \implies \text{SigSpan} \subseteq \text{span}(V_B) \subseteq \text{rank}(V_A) \leq n_B$$

For any fixed $V_B$ , each stochastic gradient $\nabla_{v_i} l$ is a sum of a vector in $\text{span}(V_B)$, i.e., the *signal part* and the weight-decay term $\lambda v_i$. Therefore, the *directional signal* that updates $v_i$ lives *inside* $\text{span}(V_B)$;

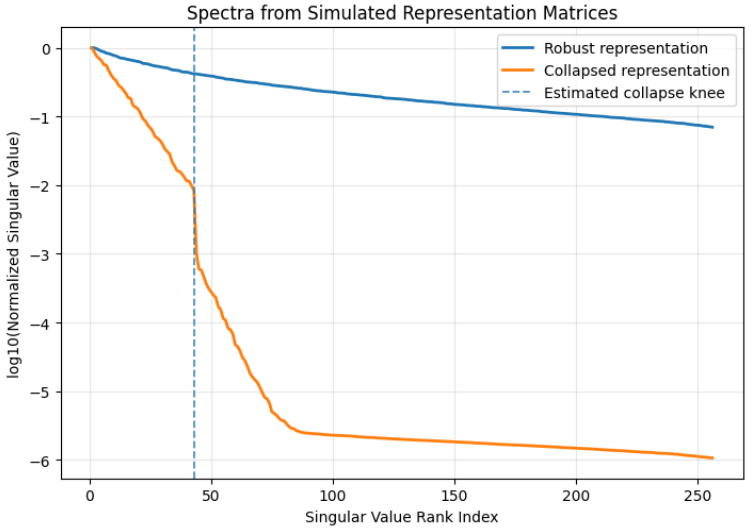
Understanding Dimensional Collapse in Contrastive Self-Supervised Learning. ICLR, 2022.
On the Embedding Collapse When Scaling Up Recommendation Models. ICML 2024.

# Dimensional Robustness

## Singular Spectrum Analysis

$$E = U \Sigma V^{\top}$$

$E$ is the **embedding matrix**, it can be both sample-wise or feature-ID-wise. After **SVD**, The singular values $\Sigma$ reflects the **scaling in each space dimension**.




## Dimensional Collapse

Definition: the embedding vectors end up spanning a **lower-dimensional subspace** instead of the entire available space

## Information Abundance metric

$$IA(E) = \frac{\sum_i \sigma_i(E)}{\max_i \sigma_i(E)} = \frac{||\sigma(E)||_1}{||\sigma(E)||_\infty}$$

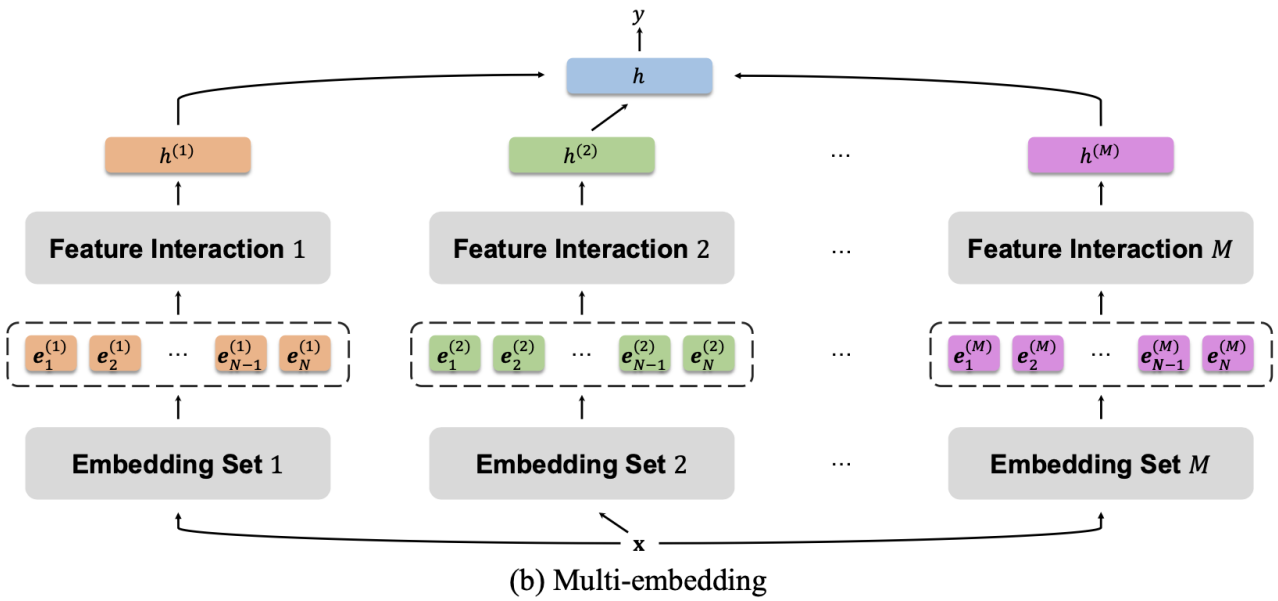
Singular sum divided by the max of singular value. The more smooth of the spectrum, the higher of IA.


Understanding Dimensional Collapse in Contrastive Self-Supervised Learning. ICLR, 2022.
On the Embedding Collapse When Scaling Up Recommendation Models. ICML 2024.

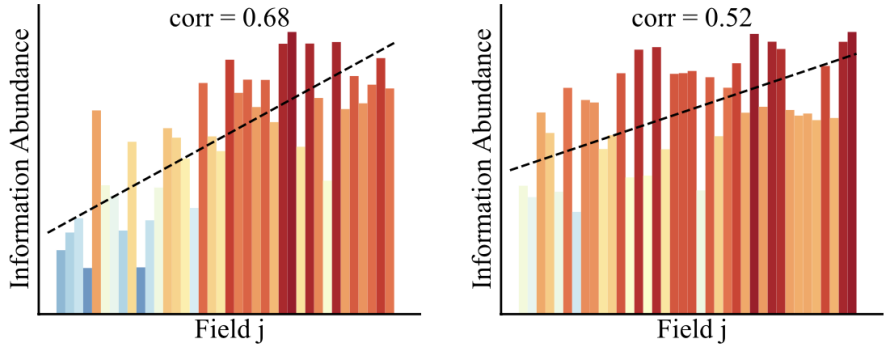
# Escaping from Dimensional Collapse

# Approach I: Multi-Embedding Paradigm

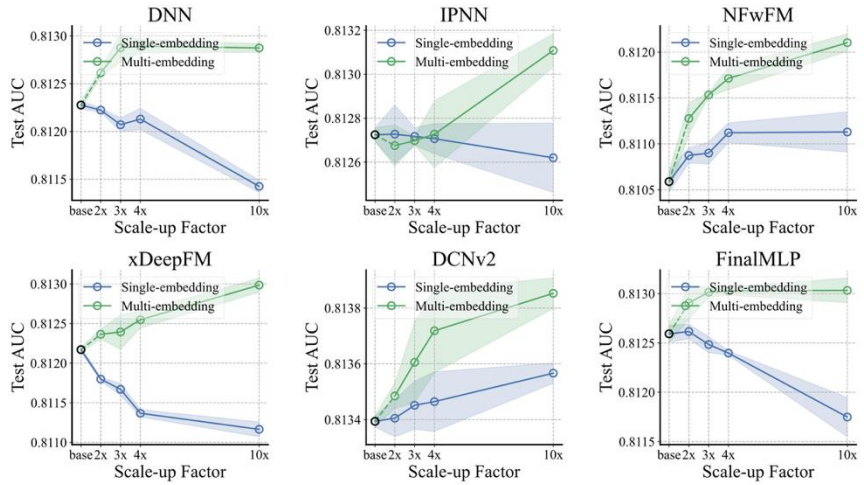
Instead to learn **a long and collapse-prone embedding** for each feature, we learn **multiple short yet dimensionally robust embeddings**.




(b) Multi-embedding




(a) $\sum_{i=1}^{N} \mathrm{IA}(\boldsymbol{E}_i^{\to j})$, SE. (b) $\sum_{i=1}^{N} \mathrm{IA}(\boldsymbol{E}_i^{\to j})$, ME.




On the Embedding Collapse When Scaling Up Recommendation Models. ICML 2024.

# Approach II: Non-Linear Interaction

Traditional feature interaction model interacts/multiplies two vectors **directly**, which make the collapse easy to **propagate through gradient**.

$$\Phi_{\mathrm{FM}} = \sum_i \sum_j v_i \odot \boldsymbol{v}_j$$

We find that, after we employ **a projection** $W$ and **non-linear activation** $\sigma(\cdot)$, the dimensional collapse can be largely mitigated:

$$\Phi_{\sigma-\mathrm{FM}} = \sum_j \sigma([v_i] \cdot W) \odot \boldsymbol{v}_j$$

Assume A is a high-cardinality feature and B is a low-cardinality one.

Vanilla FM

$$v_i \in A, v_j \in B, W = \text{projection matrix}$$

$$\nabla_{v_i} l = (\sigma(s) - y) v_j + \lambda v_i \implies \mathrm{SigSpan} \subseteq \mathrm{span}(V_B) \subseteq \mathrm{rank}(V_A) \leq n_B$$

In vanilla FM, the signal span of the high-cardinality feature lies in a low-dimension space span by the low-cardinality features

Non-Linear FM

$$\nabla_{v_i} l = (\sigma(s) - y) W^{\top} J_i v_j + \lambda v_i, \qquad J_i = Diag(\phi'(W v_i))$$

$$\mathrm{span}\{J_i v_j\} = R^r \implies \mathrm{SigSpan} = \mathrm{im}\,(W^{\top}) \implies \mathrm{rank}(V_A) = \mathrm{rank}(W)$$

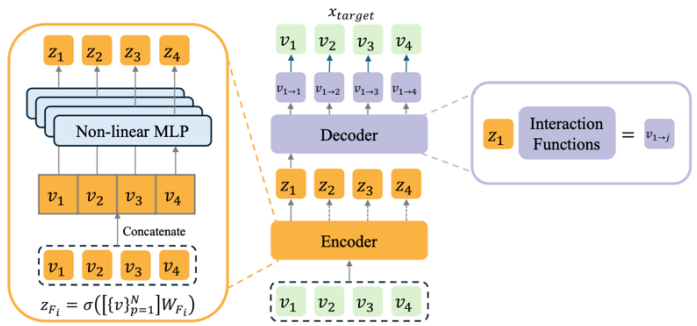
so the *signal part* of the gradient lies in the fixed subspace $S_p = \mathrm{im}(W^{\top})$, independent of $\mathrm{span}(V_B)$. Crucially, the Jacobian $J_i$ depends on $v_i$ and varies across features $i$, which removes the fixed-span bottleneck induced by the small B-table.

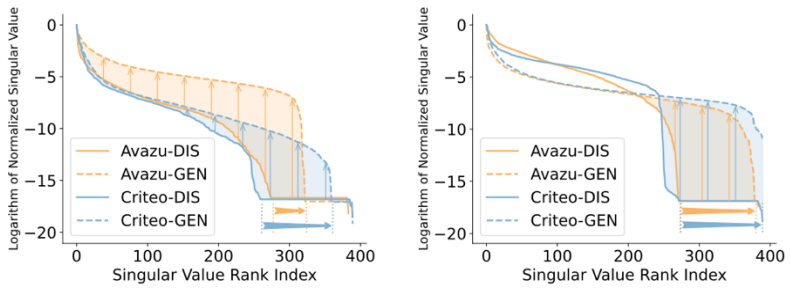
From Feature Interaction to Feature Generation: A Generative Paradigm of CTR Prediction Models. ICML, 2025.

# Approach II: Non-Linear Interaction

Non-Linear Interaction can **be adapted to almost all existing feature interaction models**, to replace the existing inner-dot or element-wise multiplication.

| Model | Discriminative | Generative |
|---|---|---|
| FM | $\sum_{i,j=1}^{N} \boldsymbol{v}_j \odot \boldsymbol{v}_i$ | $\sum_{i,j=1}^{N} \boldsymbol{\sigma}([\boldsymbol{v}] \cdot W_{F(j)}) \odot \boldsymbol{v}_i$ |
| FmFM | $\sum_{i,j=1}^{N} \boldsymbol{v}_j \odot [\boldsymbol{v}_i \cdot M_{F(i)\to F(j)}]$ | $\sum_{i,j=1}^{N} \boldsymbol{\sigma}([\boldsymbol{v}] \cdot W_{F(j)}) \odot [\boldsymbol{v}_i \cdot M_{F(i)\to F(j)}]$ |
| CrossNet V2 | $\sum_{l=1}^{L}\sum_{i,j=1}^{N} \boldsymbol{v}_j^0 \odot (\boldsymbol{v}_i^l \cdot M_{F(i)\to F(j)}^l)$ | $\sum_{l=1}^{L}\sum_{i,j=1}^{N} \boldsymbol{\sigma}([\boldsymbol{v}]^l \cdot W_{F(j)}^l) \odot (\boldsymbol{v}_i^l \cdot M_{F(i)\to F(j)}^l)$ |
| DeepFM | $\sum_{i,j=1}^{N} \boldsymbol{v}_j \odot \boldsymbol{v}_i + \mathrm{DNN}([\boldsymbol{v}])$ | $\sum_{i,j=1}^{N} \boldsymbol{\sigma}([\boldsymbol{v}] \cdot W_{F(j)}) \odot \boldsymbol{v}_i$ |
| xDeepFM | $\sum_{l=1}^{L}\sum_{i,j=1}^{N} \mathrm{Conv}^l(\boldsymbol{v}_j^0 \odot \boldsymbol{v}_i^l) + \mathrm{DNN}([\boldsymbol{v}])$ | $\sum_{l=1}^{L}\sum_{i,j=1}^{N} \mathrm{Conv}^l(\boldsymbol{\sigma}([\boldsymbol{v}]^l \cdot W_{F(j)}^l) \odot \boldsymbol{v}_i^l) + \mathrm{DNN}([\boldsymbol{v}])$ |
| IPNN | $\mathrm{DNN}([[\boldsymbol{v}], \sum_{i,j=1}^{N} \boldsymbol{v}_j \odot \boldsymbol{v}_i])$ | $\mathrm{DNN}([[\boldsymbol{v}], \sum_{i,j=1}^{N} \boldsymbol{\sigma}([\boldsymbol{v}] \cdot W_{F(j)}) \odot \boldsymbol{v}_i])$ |
| DCN V2 | $\sum_{l=1}^{L}\sum_{i,j=1}^{N} \boldsymbol{v}_j^0 \odot (\boldsymbol{v}_i^l \cdot M_{F(i)\to F(j)}^l) + \mathrm{DNN}([\boldsymbol{v}])$ | $\sum_{l=1}^{L}\sum_{i,j=1}^{N} \boldsymbol{\sigma}([\boldsymbol{v}]^l \cdot W_{F(j)}^l) \odot (\boldsymbol{v}_i^l \cdot M_{F(i)\to F(j)}^l) + \mathrm{DNN}([\boldsymbol{v}])$ |



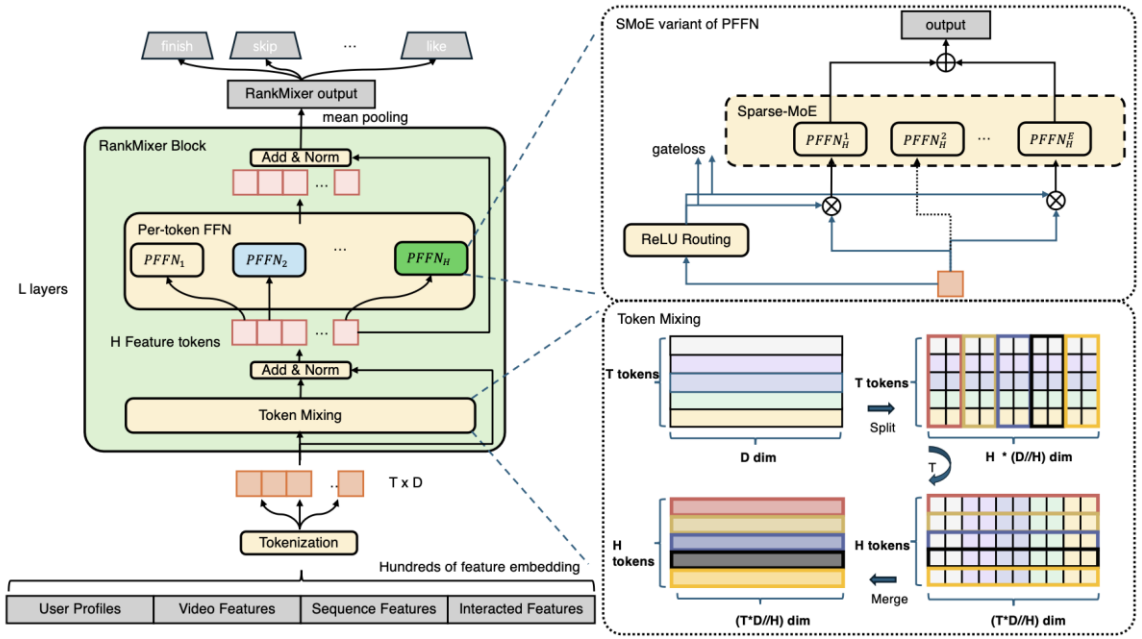



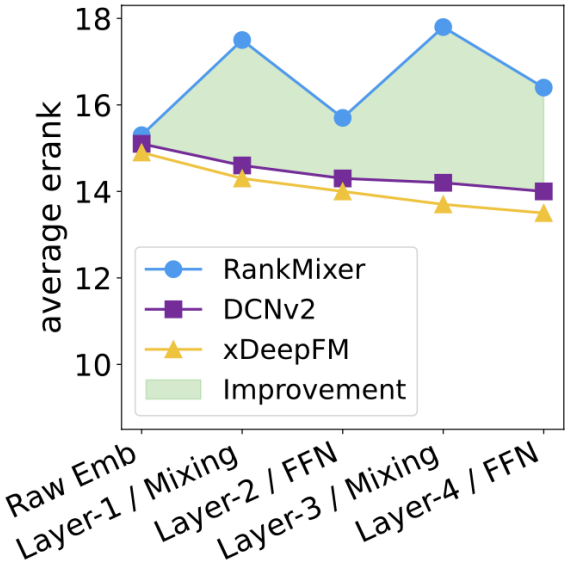

(a) FM (b) DeepFM

From Feature Interaction to Feature Generation: A Generative Paradigm of CTR Prediction Models. ICML, 2025.

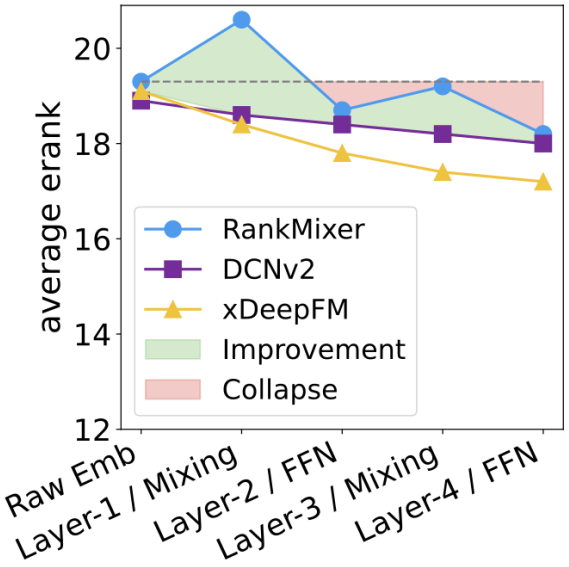
# Approach III: TokenMixing

- **TokenMixing**:
  - Token Split
  - Token Mix
- **Per-token FFN**:
  - A Gelu non-linear activation




**(a) Comparison on Criteo.**




**(b) Comparison on Avazu.**

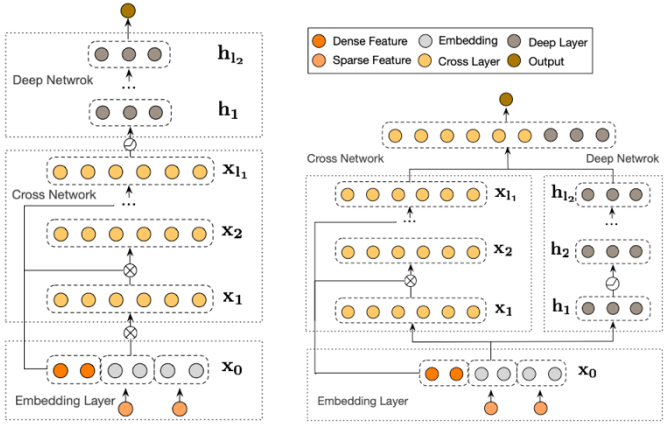
- There is a monotonic dimensional collapse trend along layer
- RankMixer present a “sawtooth” pattern:
  - Mixing can boost dimensional robustness
  - P-FNN reduce the effective dimension

RankMixer: Scaling Up Ranking Models in Industrial Recommenders. CIKM, 2025.
Expand More, Shrink Less/ Shaping Effective-Rank Dynamics for Dense Scaling in Recommendation. SIGKDD 2026.

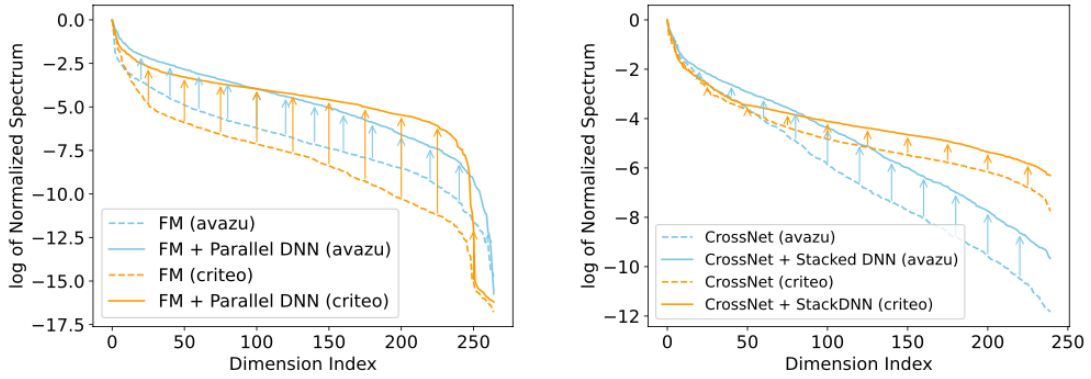
# Approach IV: DNN

- Why hybrid models work better?
- Does DNN really learn implicit interaction?
- Rendle proves that **DNN is hard to learn dot products**.

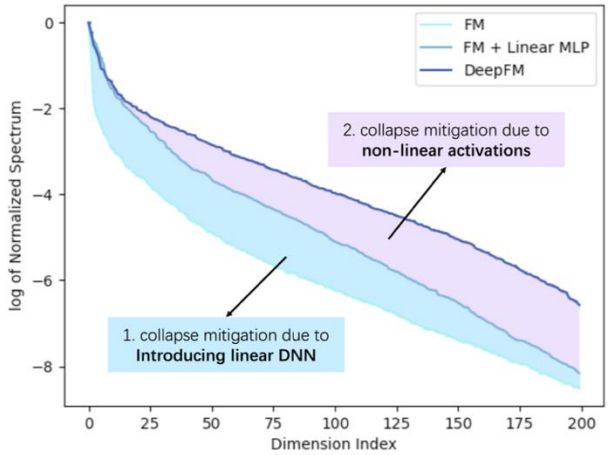
The **parallel and stacked DNN** mitigates the dimensional collapse of embeddings in recommendation models.




(a) Stacked (b) Parallel




(a) Parallel DNN (b) Stacked DNN




Both **linear DNN** and **non-linear activations** can mitigate **dimensional collapse**.


Neural Collaborative Filtering vs. Matrix Factorization Revisited. RecSys 2020.
Understanding DNNs in Feature Interaction Models: A Dimensional Collapse Perspective. SIGIR 2026..

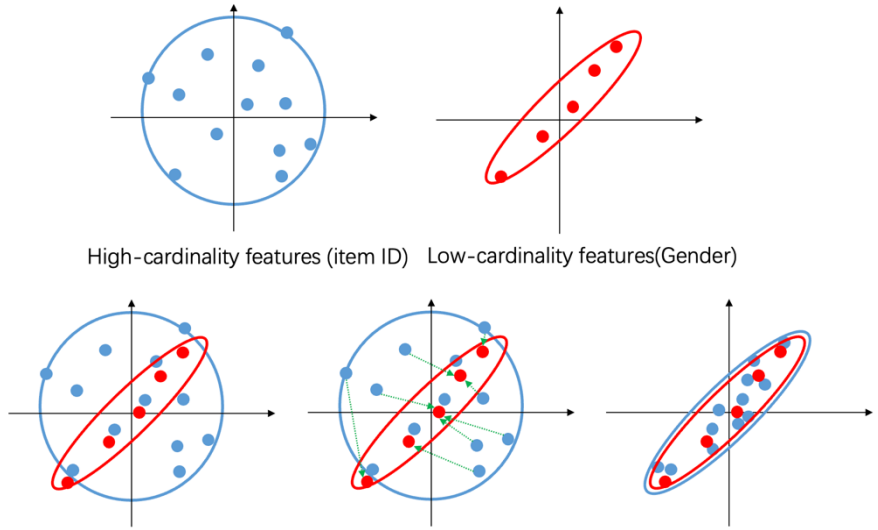
# Main Takeaways

In feature-interaction models:

- There are some **dimensionally-ill features**, such as low-cardinality features (gender).
- Due to **interaction-collapse theory**, these features will **make the embeddings of other features collapse**.

The community has proposed various methods to mitigate the dimensional collapse:

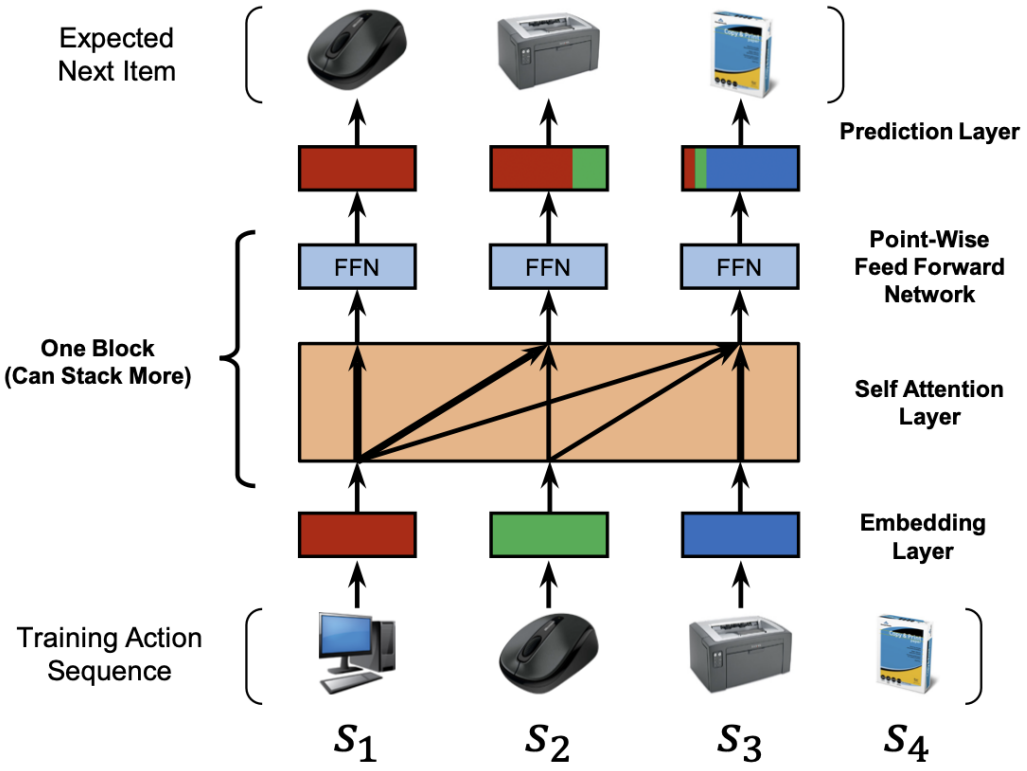
- **Multi-Embedding**
- **Non-linear Interaction**: Feature Generation
- **TokenMixing**: RankMixer
- **DNN**: Wide & Deep, DeepFM, DCN V2



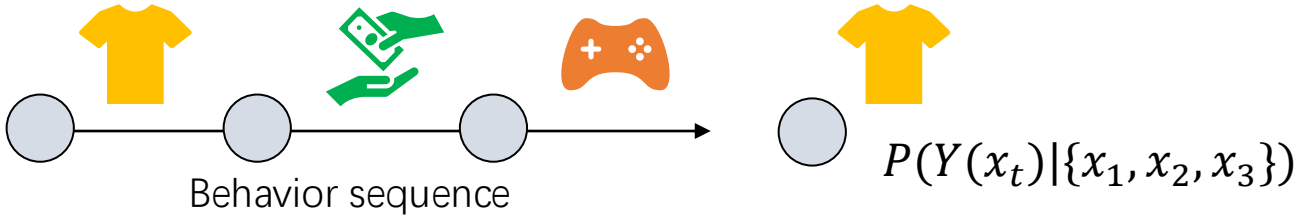
# Sequential Recommender

# SASRec & DIN

- The first work to employ Transformer/Self-Attention for recommendation.
- Formulate a next-token prediction problem.

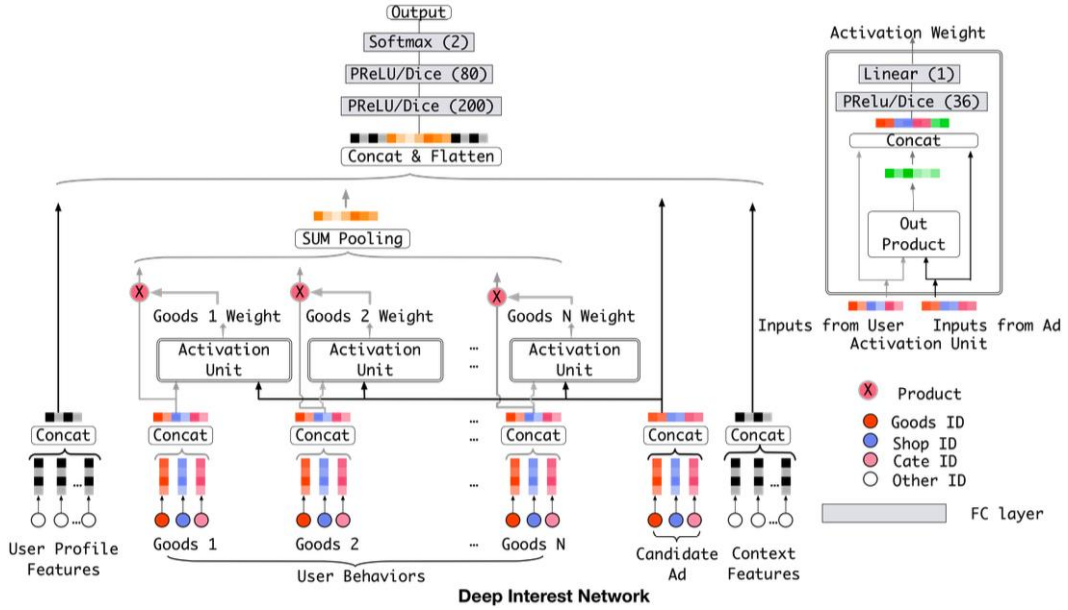



Self-Attentive Sequential Recommendation. ICDM, 2018.




In prediction, the most import sequential correlation is the behavior-target correlation.

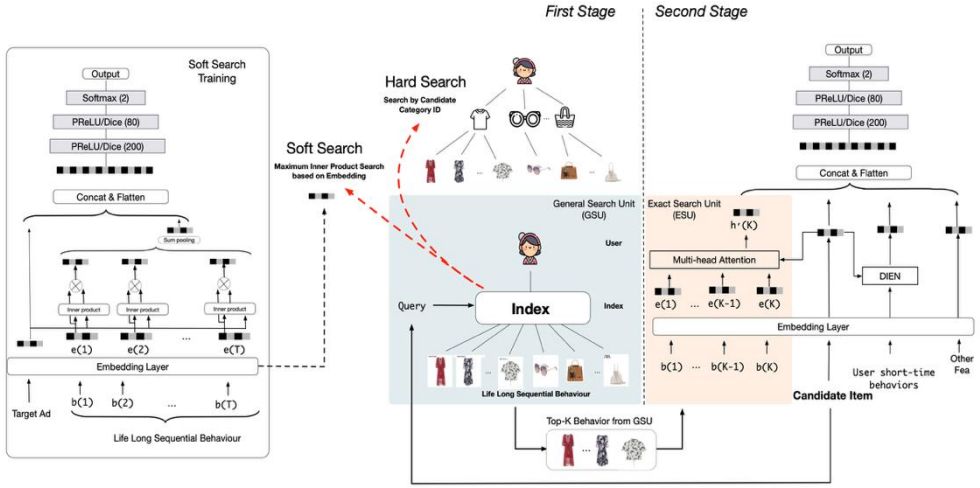
DIN: **Target Attention**, captures the correlation between each **behavior** and the **target**.



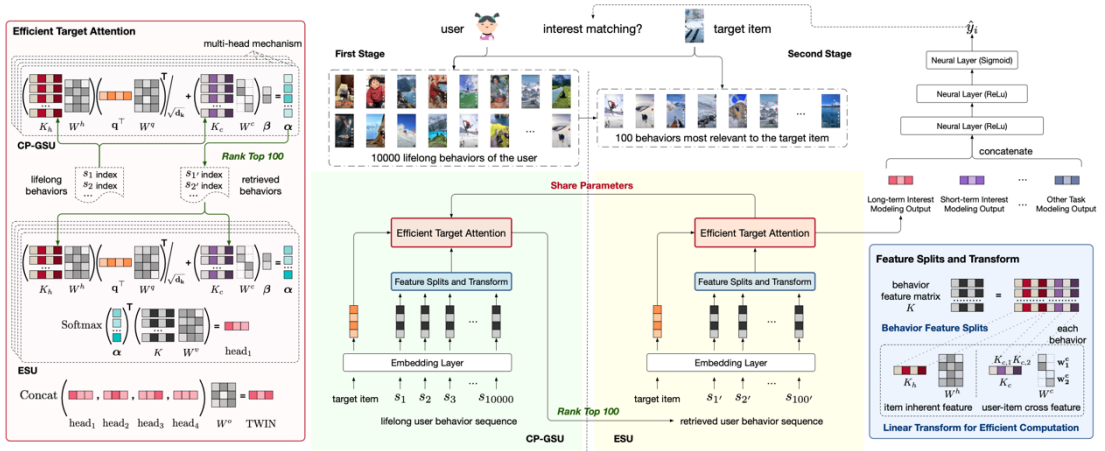

Deep Interest Network for Click-Through Rate Prediction. KDD 2018.

# Long-Sequence Modeling

- User's behavior sequence with length of 1K ~ 1M, making **target-attention** infeasible
- Two-stage design for efficient user interest modeling
  - Stage 1: General Search Unit (**GSU**): retrieve the most informative behaviors
    - **Hard Search**: Same category,.
    - **Soft Search**: Attention score.
  - Stage 2: Exact Search Unit (**ESU**): model user interest on the retrieved behaviors, e.g., DIN.




SIM




DCN V2

Search-based User Interest Modeling with Lifelong Sequential Behavior Data for Click-Through Rate Prediction. CIKM 2020.
TWIN: TWo-stage Interest Network for Lifelong User Behavior Modeling in CTR Prediction at Kuaishou. KDD 2023.
TWIN V2: Scaling Ultra-Long User Behavior Sequence Modeling for Enhanced CTR Prediction at Kuaishou. CIKM 2024.

# Revisit Dimensional Collapse in Sequential Models

KDD2026 August 9–13, 2026 Jeju, Korea
Tencent Marketing Solution

In feature-interaction models:

- There are some **dimensionally-ill features**, such as low-cardinality features (gender).
- Due to interaction-collapse theory, these features will **make the embeddings of other features collapse**.

In sequential models:

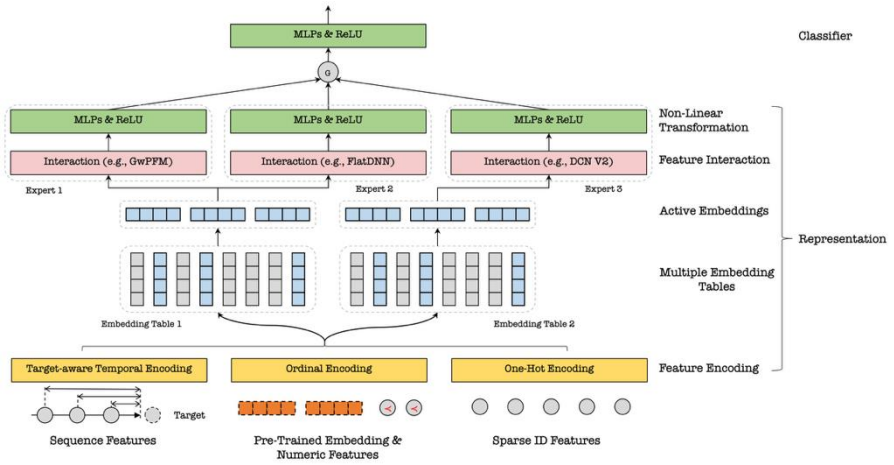
- The interaction (attention) mainly happens between the behaviors within the sequence
- There **are fewer dimensionally-ill features** in sequence, since we mainly use category or item IDs as side-info.
- Therefore**, there is no dimensional collapse issue** in sequential recommendation models

# Bridging the Two Worlds

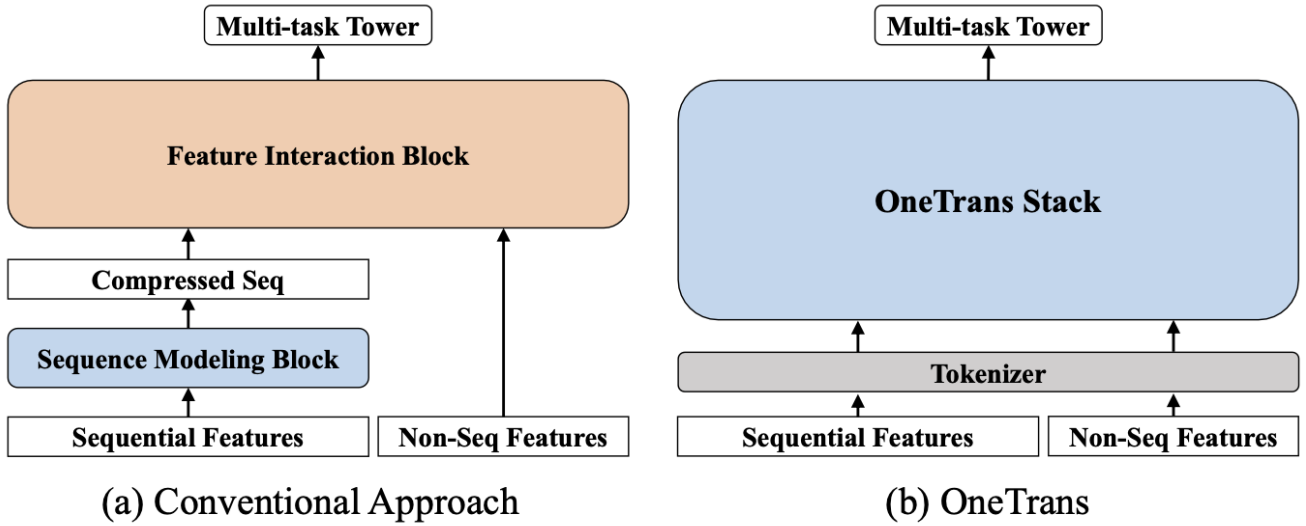
# OneTrans

Core motivation:

- Employ unified blocks for sequential and non-sequential modeling
- The entire model can be scaled by adjusting backbone depth and width



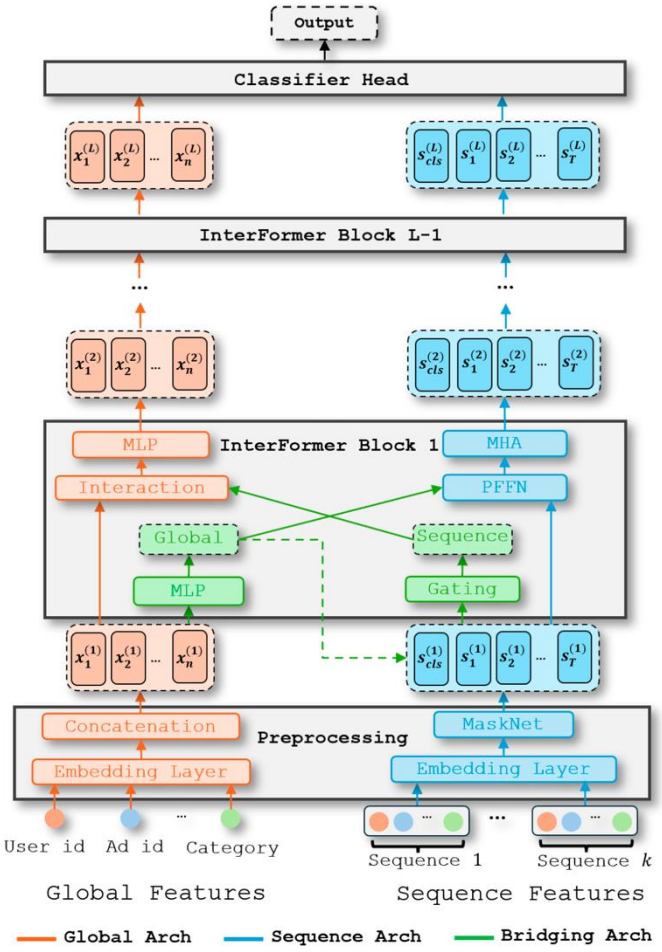

Our production model at 2024



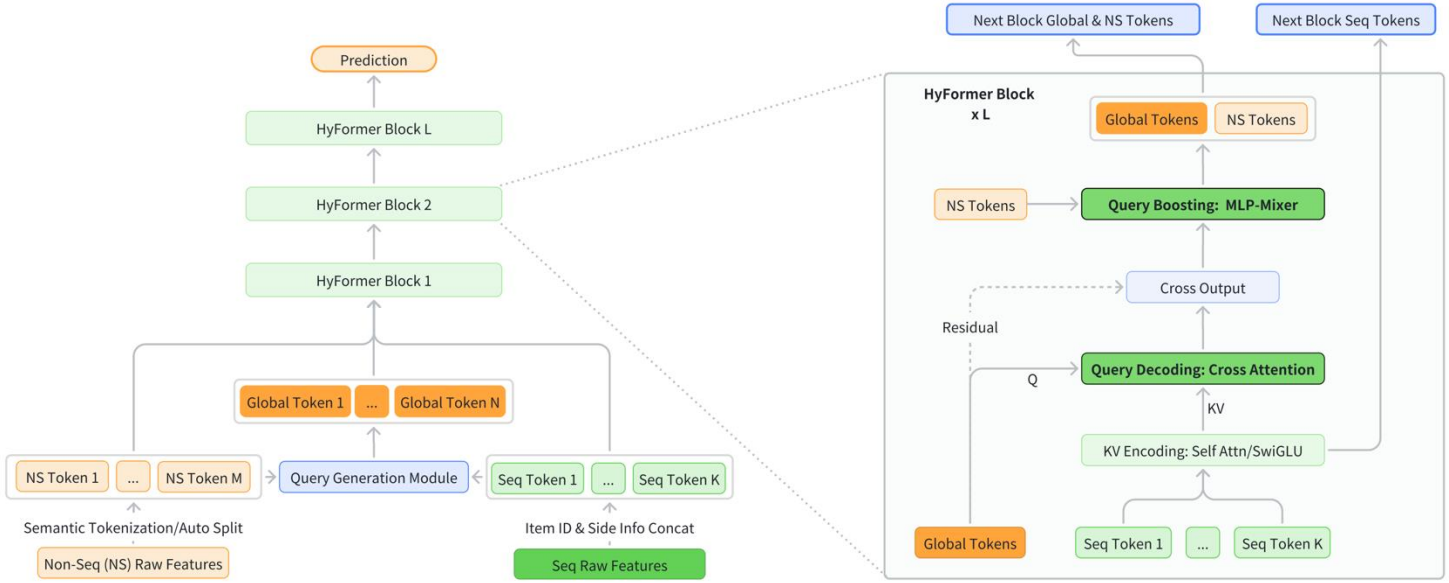

OneTrans

OneTrans: Unified Feature Interaction and Sequence Modeling with One Transformer in Industrial Recommender. WWW, 2026.

# InterFormer & HyFormer



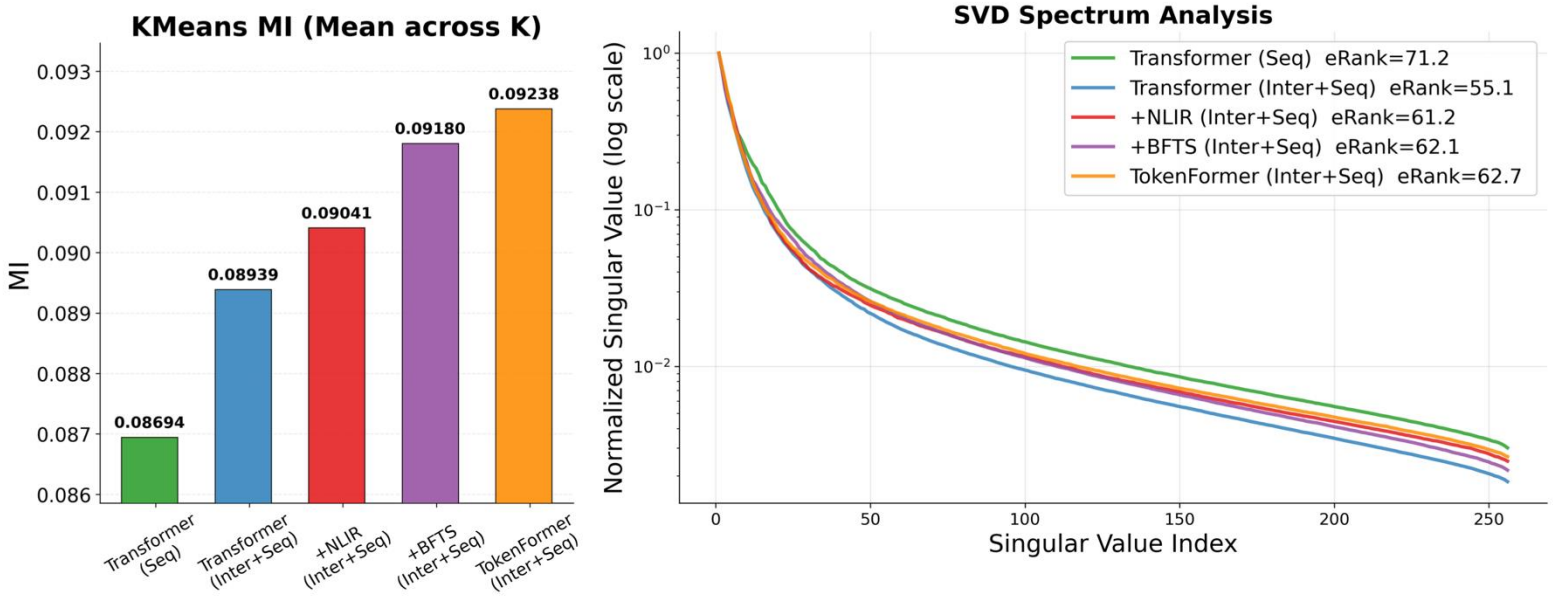

InterFormer




HyFormer

InterFormer: Towards Effective Heterogeneous Interaction Learning for Click-Through Rate Prediction. CIKM, 2025.
HyFormer: Revisiting the Roles of Sequence Modeling and Feature Interaction in CTR Prediction. Arxiv, 2026.

# Sequential Collapse Propagation




- By incorporating the non-sequence features, we **get a higher discriminability**.
- However, there is significant collapse of the singular spectrum of the sequential tokens, leading to a **Sequential Collapse Propagation** challenge.

TokenFormer- Unify the Multi-Field and Sequential Recommendation Worlds. Arxiv, 2026.

# Revisit Dimensional Collapse in Unified Models



In **Feature-Interaction** models:

- There are some **dimensionally-ill features**, such as low-cardinality features (gender).
- Due to interaction-collapse theory, these features will **make the embeddings of other features collapse**.

In **Sequential** models:

- The interaction (attention) mainly happens between the behaviors within the sequence
- There **are fewer dimensionally-ill features** in sequence, since we mainly use category or item IDs as side-info.
- Therefore, **there is no dimensional collapse issue** in sequential recommendation models

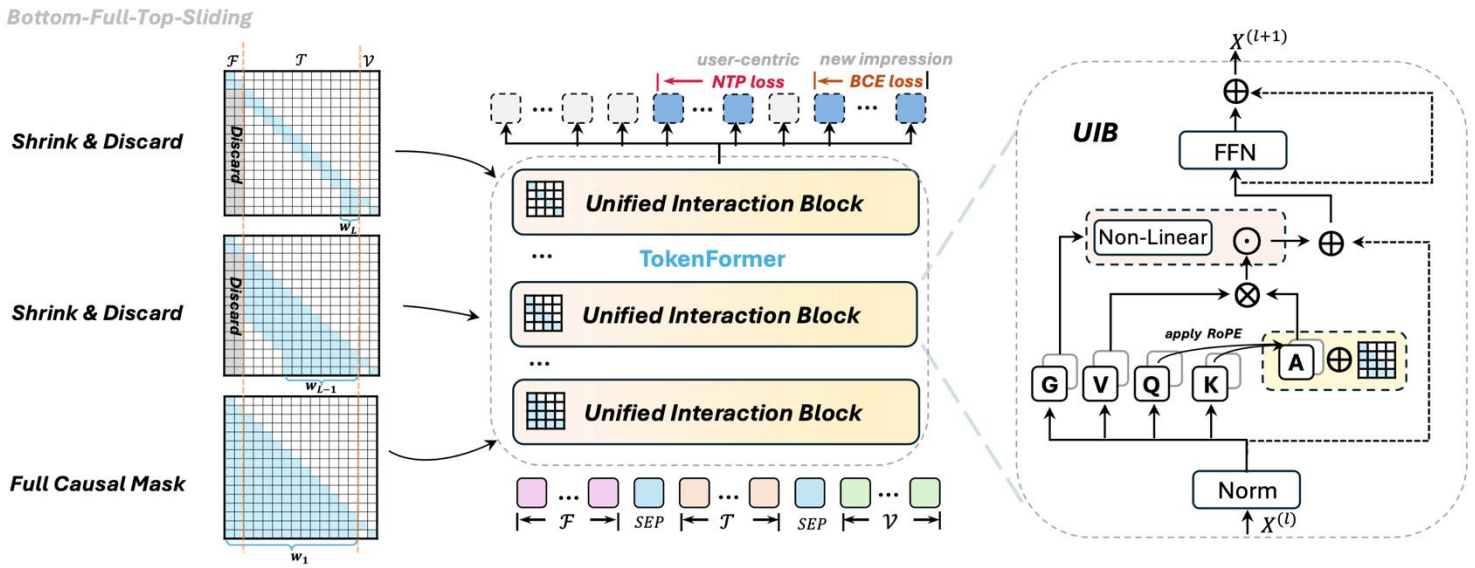
In **Unified** models:

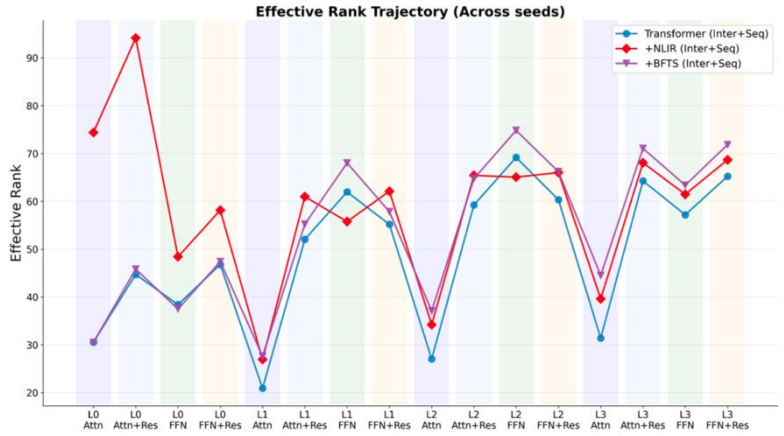
- There are interaction between **dimensionally-ill non-sequence features** and sequence features
- The interaction with those **dimensionally-ill non-sequence fields** leads to the **dimensional collapse** of the sequence features, i.e., the **Sequential Collapse propagation** happens.

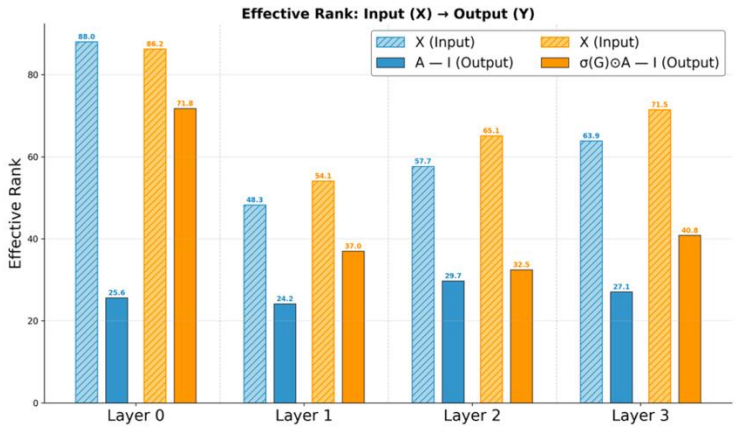
# TokenFormer

Key approaches to mitigate dimensional collapse:

- **Bottom-Full-Top-Sliding** attention schedule
- **Non-Linear Interaction Representations**



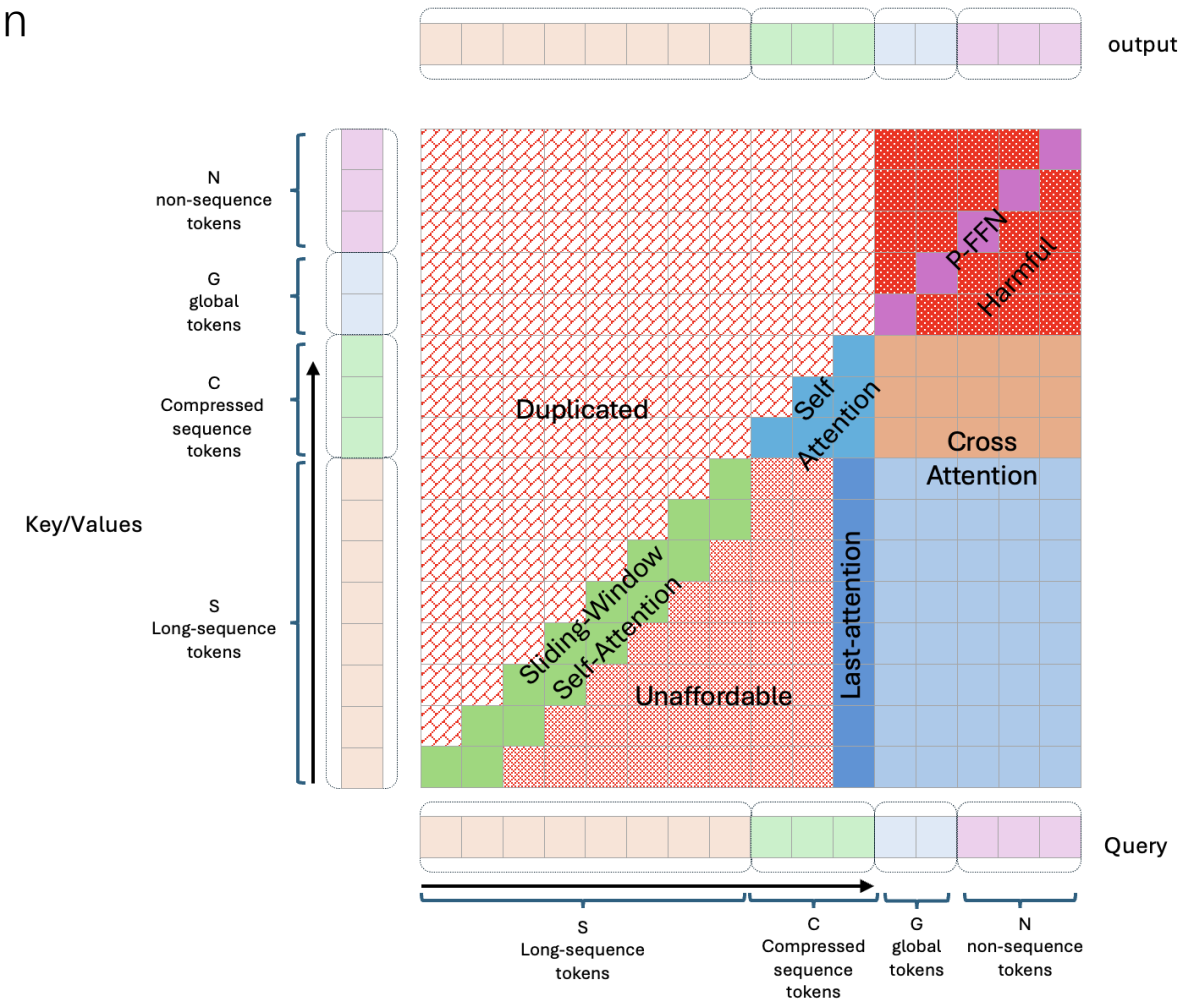

TokenFormer- Unify the Multi-Field and Sequential Recommendation Worlds. Arxiv, 2026.

# TokenFormer

Features can be **transformed to tokens** in recommendation:

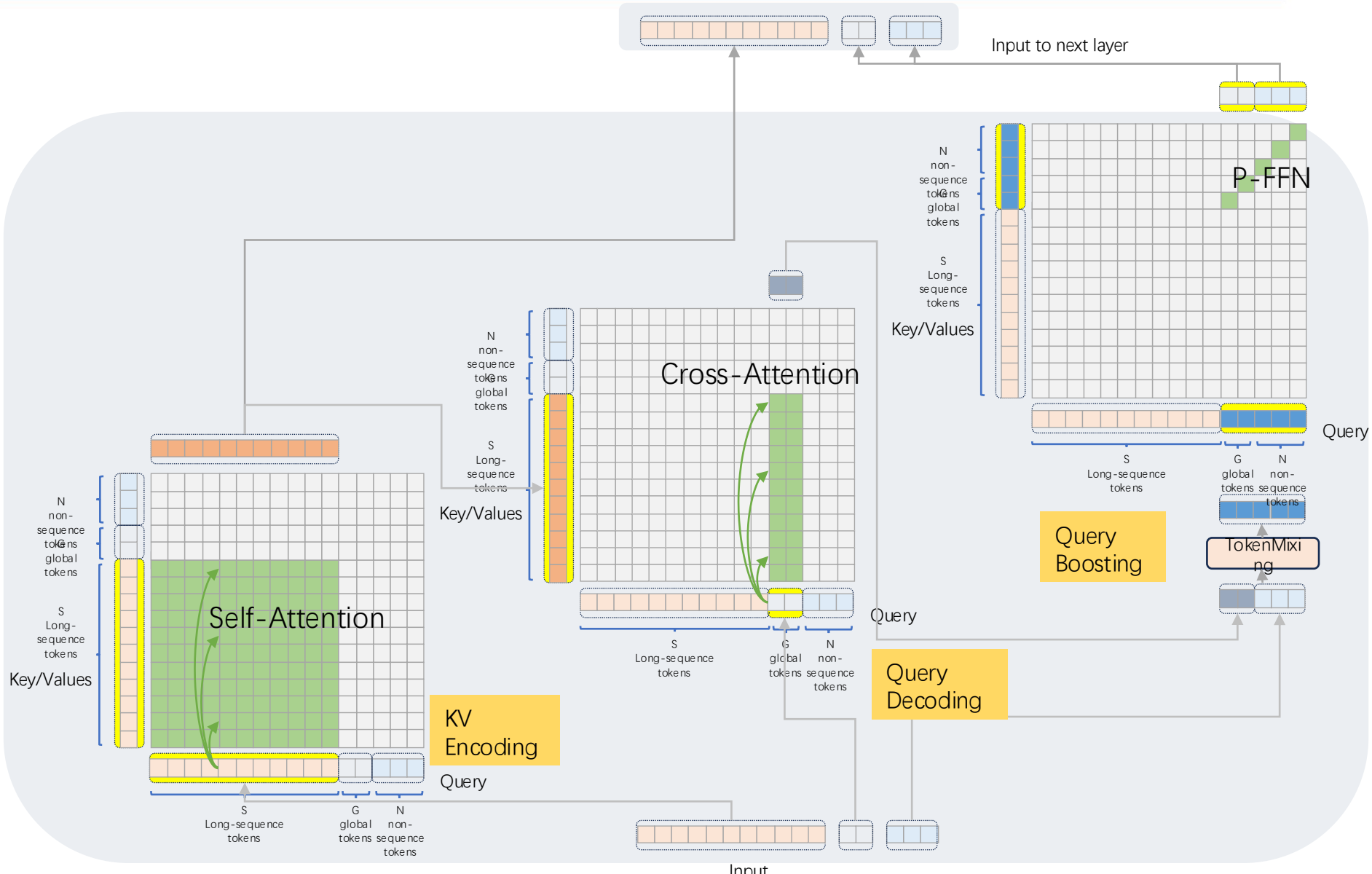
- Non-Sequence Tokens
- Global Tokens
- Raw Sequence Tokens
- Compressed Sequence Tokens

The interactions between all tokens of all types can be represented as the right "**interaction map**", which can be implemented by transformer **attention masks**.




TokenFormer- Unify the Multi-Field and Sequential Recommendation Worlds. Arxiv, 2026.

# HyFormer under view of Interaction Map



# KDD Cup x TAAC 2026

# KDD Cup x Tencent Advertising Algorithm Competition

KDD2026 August 9–13, 2026 Jeju, Korea

Tencent Marketing Solution

Towards Unifying Sequence Modeling and Feature Interaction for Large-scale Recommendation

| Academic Track | Total $540,000 |
|---|---|
| Champion (1 team) | $300,000 |
| Runner-up (1 team) | $90,000 |
| Third Place (1 team) | $45,000 |
| Honorable Mentions (rank4–10, 7 teams) | Per team $15,000 |

| Industrial Track | Total $255,000 |
|---|---|
| Champion (1 team) | $150,000 |
| Runner-up (1 team) | $75,000 |
| Third Place (1 team) | $30,000 |
| Honorable Mentions (rank4–10, 7 teams) | Each participating team member stands to win an Apple iPad Air |

- Unified Block Innovation Award
- Scaling Law Innovation Award

# KDD Cup x Tencent Advertising Algorithm Competition

KDD2026 August 9–13, 2026 Jeju, Korea

Tencent Marketing Solution

## Across 52 Countries and Regions

## 13,913 Participants

### Competition Snapshot

- **71%** Academia
- **29%** Industry
- **5746** Total Teams
- **657** Mainland Universities
- **20** Hong Kong, Macao and Taiwan Universities
- **230** Overseas Universities

TAAC 腾讯广告算法大赛 KDD2026

UniRec

TAAC × KDD Cup 2026

## Beyond the Glory A Journey Ahead

AUG 10 — TOP Revealed

AUG 12 — Workshop · Frontier Tech Exchange

### Workshop Highlights

13:30-17:30 Keynotes & Winning Solutions Premiere

**Academic Pioneer**

Irwin King
Professor, Vice-President, CUHK

**Industrial Expert**

Rui Li
Principal AI Research Scientist @Meta

Fedor Borisyuk
Principal Staff Software Engineer @LinkedIn

**Winning Solutions Premiere**

Showcase of champion solutions from both the Academic and Industrial Tracks.

17:30-17:40 Award Ceremony & Closing Remarks

17:40 Poster Session

18:30 Dinner (venue TBD)

From the arena to the stage, a feast of ideas on Ad-tech evolution.

TIME: August 12, 13:30 (KST)

LOCATION: ICC-1, Room 303B



# Thanks!

# Q & A